



Artistic bacteria

With genetically engineered colour vision, MIT researchers are transforming bacteria into vibrant artists. <http://digit.in/bacteria3>



Vulnerable nuclear subs

Local researchers warn of possible WannaCry infection as Britain's nuclear submarines are still on Windows XP. <http://digit.in/wannaNuke>

AMD Ryzen 5 1500X

More mainstream madness

AMD Ryzen 7 1800X really did a number on the PC market. CPU prices came tumbling down with the entry of an 8 core / 16 thread CPU that nearly equalled multi-threaded performance of processors that cost twice as much. The fledgling platform is still dealing with a few teething problems with regards to single-threaded applications but there's hope. The 1500X has an MRP of ₹13,799, but you really can't find it anywhere for that price. ₹14,200 is the lowest we've spotted it and for that, the competition is the Intel Core i5 - 7500 – a 4 core / 4 thread CPU that's become a favourite for mid-range gaming machines.



Price
₹13,799



PERFORMANCE.....74
VALUE FOR MONEY.....90

We pitted the 1500X against the Core i5 - 7500 and found it to be very competitive. Handbrake video transcoding was 42% faster than the 7500 and in Cinebench R15, it scored 155 which was 1 point lower than the 7500 in single threaded test but in the multi-threaded test the 1500X managed a 31%

lead. So we're still going to see multi-media applications executing much faster on the Ryzen machine provided the software optimisations favour the Ryzen architecture. Coming to gaming, there's an interesting pattern that's visible. When the same title is compared at medium to high settings, the performance is similar but the moment you tone things down to minimum, the Intel based system scores a significant lead. Overall, there's need for optimisation with regards to game engines and memory as well but with those things kept aside, the Ryzen 5 1500X makes for a really great mid-range CPU.

—Mithun Mohandas

SPECIFICATIONS

SOCKET: AM4 | CORES: 4 | THREADS: 8 | BASE CLOCK: 3.5 GHz | BOOST CLOCK: 3.7 GHz | XFR: 3.9 GHz | L2 CACHE: 2 MB | PROCESS NODE: 14nm

CONTACT

AMD | PHONE: NA | SUPPORT: support.amd.com | WEBSITE: www.amd.com

HP Omen 17 Gaming Laptop

Late to the party

We'll start off by saying there was a disparity between the specifications of the unit that we received and that one that is available in the Indian market. Our review unit has an Intel Core i7 6700HQ and a 4K display panel while the Indian unit has an Intel Core i7 7700HQ and a Full HD panel. In a way, you are getting a better performing CPU but you miss out on a high-res display. The HP Omen 17 comes in a backpack with HP Omen branded gaming mouse and headset manufactured by Steel-Series. Good choice, but we couldn't help but notice that HP is a bit late to the gaming party in India.

The laptop has a pretty slick looking textured design that not only adorns



Price
₹1,69,990



PERFORMANCE.....81
FEATURES.....78
BUILD.....80

the back of the display panel but also extends to the chassis front. Inside it has dual cooling assemblies for the CPU and the GPU. The access hatch only opens up to the RAM so replacing the HDD and SSD will require opening the entire cover. Performance wise, the HP Omen 17 does take a few steps back because of the older

CPU but when it comes to gaming it decimates practically most of the competition. With the 1070, 4K gaming is pretty easily accomplished on the unit, however, since the Indian units are limited to just FullHD, the FPS counter is going to be riding high in the double digits for a year or two. There aren't that many GTX 1080 laptops and even fewer GTX 1070 laptops, so HP's bagged itself a pretty good configuration and price point to sell the Omen 17.

—Mithun Mohandas

SPECIFICATIONS

PROCESSOR: Intel Core i7 6700HQ | RAM: 16 GB 2400 MHz DDR4 | SCREEN SIZE: 17.3-inches | SCREEN RESOLUTION: 3840x2160 | GPU: GTX 1070 8GB | STORAGE: 512GB M.2 NVMe SSD + 1TB HDD | ETHERNET: Gigabit Ethernet | BATTERY: 6-Cell 95.8 Whr | AC ADAPTER: 230 W

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